

Pranavsai Gandikota

954-282-9164 | pranavsagandikota@gmail.com | linkedin.com/in/pranavsai | github.com/pranavsagandikota | pranavsai.dev

EDUCATION

University of Central Florida

Orlando, FL

B.S. Computer Science | **GPA: 3.99/4.0** | President's Honor Roll 2024, 2025 | Dean's List 2026

Aug. 2024 – May 2028

Activities: SASE (CS Tech Chair), KnightHacks (Web Dev/Design), AI UCF (Projects Director), SHPE

EXPERIENCE

Software Engineering Intern

May 2026 – Aug. 2026

Ford Motor Company (Global Trade Services, Order Fulfillment, Marketing & Sales Technology)

Dearborn, MI

- Modernized 6 legacy **Struts** batch jobs to a **Spring Boot** microservice on **Google Cloud Run**, exposing **REST** endpoints that trigger **PostgreSQL** stored procedures; reduced average processing time by **92%** from 45 minutes to 3.5 minutes while preserving byte-exact fixed-width output for a **SOX**-compliant downstream invoicing system.
- Validated migrated extracts against **200K+** records and coordinated SFTP delivery with overseas partner teams to ensure downstream invoicing workflows continued without functional changes.
- Engineered an **event-driven Pub/Sub/Cloud Functions** pipeline that automatically refreshes affected **Spring Boot** services through **Apigee** after **Secret Manager** rotations using dynamically generated **Azure AD (Entra ID)** tokens.
- Replaced a **3–7 hour** manual restart and redeploy process during large incidents with automated service refreshes across **84 services and 49 secrets**, using **Cloud Storage** JSON mappings to onboard new services without code changes.
- Presented the service-refresh automation to engineering leaders and **109 engineers** in an internal software craft session.
- Developed an **NL2SQL** solution that translated natural-language business questions into **SQL** queries over relational data.

Teaching Assistant – Object-Oriented Programming with Java

Jan. 2026 – May 2026

College of Engineering and Computer Science, UCF

Orlando, FL

- Graded assignments and exams for correctness and algorithmic efficiency, and held weekly office hours supporting **166** students on **Java** OOP design, structured programming, and debugging.

Research Assistant

Sep. 2024 – Dec. 2025

Interactive Systems and User Experience (ISUE) Lab, UCF

Orlando, FL

- Fine-tuned **LLaMA** with **QLoRA** to predict cognitive load and usability scores from voice replies during VR user studies.

PROJECTS

uKnight | Java, Spring Boot, WebRTC, WebSockets, Next.js, PostgreSQL, GCP Cloud Run, Docker

- Led a **6-member team** using an iterative development approach to build an anonymous video chat platform for verified .edu users, deploying the backend on **GCP Cloud Run** and reaching **72 verified users in 3 weeks**.
- Implemented real-time video and audio connections using **WebRTC**, with **WebSocket**-based signaling and **ICE**, **STUN**, and **TURN** for connection establishment and NAT traversal.

Satchel | Java, Spring Boot, Spring AI, OAuth 2.0, Auth0, PostgreSQL, React, Docker

- Built a full-stack inventory platform with **Spring Boot REST APIs** and React, implementing role-based shared inventories with **Auth0/OAuth 2.0**; containerized the backend with **Docker** and hosted the **PostgreSQL** database on **Supabase**.
- Integrated **Spring AI** with GPT to parse receipts into inventory items, track stock levels, and generate recipes based on available ingredients and expiration dates.

Vision – Knight Hacks VII | Python, OpenCV, MediaPipe, Win32 API | Honorable Mention: Best Overall

- Built an accessibility application for hands-free drawing using real-time eye tracking and voice commands; engineered a multi-threaded pipeline with direct **Win32 API** calls, reducing gaze-to-cursor latency by approximately **60%** from 160–200ms to 60–100ms.

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, TypeScript, C, SQL

Frameworks & Libraries: Spring Boot (Security, Data, AI), React, Node.js, Express, Next.js, FastAPI

Databases & APIs: PostgreSQL, MySQL, Supabase, REST APIs, OAuth 2.0, Auth0

AI / ML: PyTorch, LLaMA, QLoRA, GPT, Gemini, Pandas, NumPy, Hugging Face Transformers

Cloud & DevOps: GCP (Cloud Run, Pub/Sub, Cloud Storage, Secret Manager), Terraform, Docker, Linux, Git, CI/CD

Development Practices and Tools: TDD, Gitflow, Agile/Scrum **Tools:** Jira, Figma, GitHub